

Name:

Student ID:

Duration: 20 minutes

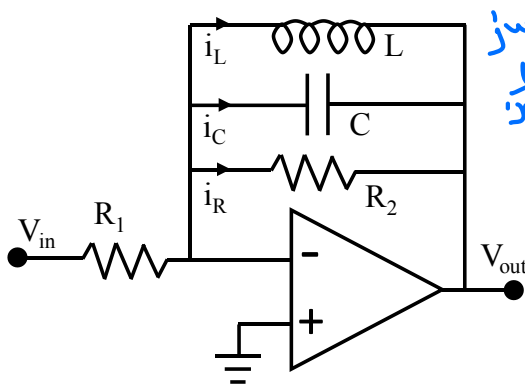
For the below circuit, assume the OPAMP is ideal without saturation, and $L=1\text{mH}$, $C=10\mu\text{F}$, $R_1=1\text{k}\Omega$, and $R_2=2\text{k}\Omega$.

a) (40 pts) Find the transfer function, $H(\omega)=V_{\text{out}}(\omega)/V_{\text{in}}(\omega)$, and resonance frequency, ω_0 (in the units of rps).

b) (20 pts) Find $v_{\text{out}}(t)$ if $v_{\text{in}}(t)=6\cos(\omega_0 t)$ V.

c) (30 pts) Find $i_L(t)$, $i_C(t)$, and $i_R(t)$ for the $v_{\text{in}}(t)$ in part b.

d) (10 pts) Calculate the quality factor (Q) using the currents you have found in part c.



$$a) \frac{V_{\text{in}}}{R_1} = -\frac{V_{\text{out}}}{j\omega L} - V_{\text{out}}(j\omega C) - \frac{V_{\text{out}}}{R_2}$$

$$\frac{V_{\text{in}}}{R_1} = -V_{\text{out}} \left(\frac{1}{j\omega L} + j\omega C + \frac{1}{R_2} \right)$$

$$H(\omega) \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{1}{R_1} \frac{1}{\frac{1}{j\omega L} + j\omega C + \frac{1}{R_2}}$$

$$H(\omega) = -\frac{1}{R_1} \frac{j\omega L R_2}{R_2 - \omega^2 L C R_2 + j\omega L} \Rightarrow H(\omega) = -\frac{1}{R_1} \frac{j\omega L R_2}{R_2(1 - \omega^2 L C) + j\omega L}$$

$$@ \omega_0 \quad (1 - \omega_0^2 L C) = 0 \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{1 \cdot 10^{-3} \cdot 10^{-5}} \Rightarrow \omega_0 = 10^4 \text{ rad/s}$$

$$b) @ \omega_0 \quad \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{1}{R_1} \frac{j\omega L R_2}{j\omega L} \Rightarrow H(\omega_0) = -\frac{R_2}{R_1}$$

$$V_{\text{out}}(t) = -2 \cdot 6 \cos(\omega_0 t) \Rightarrow V_{\text{out}}(t) = -12 \cos(\omega_0 t) \text{ V}$$

$$c) I_L = -\frac{V_{\text{out}}}{j\omega L} = \frac{12}{j 10^4 \cdot 10^{-3}} = -j 1.2 \text{ A}$$

$$I_C = -V_{\text{out}} j\omega C = 12 \cdot j 10^4 \cdot 10^{-5} = +j 1.2 \text{ A}$$

$$I_R = \frac{12}{2 \text{ k}} = 6 \text{ mA}$$

$$i_L(t) = 1.2 \cos(\omega_0 t - \pi/2) \text{ A} = 1.2 \sin(\omega_0 t) \text{ A}$$

$$i_C(t) = 1.2 \cos(\omega_0 t + \pi/2) \text{ A} = -1.2 \sin(\omega_0 t) \text{ A}$$

$$i_R(t) = 6 \cos(\omega_0 t) \text{ mA}$$

Note that

$$I_L = -I_C$$

@ ω_0

$$d) @ \omega_0 \quad \frac{|I_L| |I_C|}{I_R} = Q \quad \left| \frac{1.2}{6 \text{ mA}} \right| = Q \quad Q = 200$$